

Cloud Support Ticketing Simulation

Cloud Project | AWS Cloud Support Engineer

Tickets Created	Services Covered	Skills Demonstrated	Project Type
3	EC2, VPC, CloudFront, CI/CD	Troubleshooting, RCA, Documentation	Self-Directed

Project Overview

This project demonstrates professional Cloud Support Engineer workflows through realistic incident documentation. Each ticket follows a structured approach: incident triage, systematic investigation, root cause analysis, resolution, and preventative measures.

Learning Objectives

- Practice real-world cloud incident handling and response
- Develop systematic troubleshooting methodology
- Create professional incident documentation
- Demonstrate Cloud Support Engineer daily workflows
- Build evidence of support capabilities

Standard Incident Workflow

Step	Description
1. Incident Report	Problem identified by user or monitoring system
2. Initial Triage	Severity assessment and prioritization
3. Investigation	Systematic troubleshooting and data collection
4. Root Cause Analysis	Identify underlying cause, not just symptoms
5. Resolution	Implement fix and verify functionality
6. Preventative Actions	Document learnings and prevent recurrence
7. Documentation	Clear communication for future reference

Incident Tickets

CS-001: EC2 Instance Unreachable After Launch

Priority	High
Service	EC2, VPC
Status	Resolved
Resolution Time	1 hour 5 minutes

Problem: EC2 instance launched successfully but unreachable via SSH from client network.

Root Cause: Security group missing inbound rule for SSH (port 22) from client IP.

Resolution: Added security group rule allowing SSH access from client CIDR block (203.0.113.0/24).

Skills Demonstrated: VPC networking, security group configuration, systematic troubleshooting

Preventative Actions: Created VPC launch checklist, automated permission validation script

CS-002: CloudFront Routed to Secondary Origin Unexpectedly

Priority	Medium
Service	CloudFront
Status	Resolved
Resolution Time	1 hour 45 minutes

Problem: CloudFront failed over to backup S3 origin during routine deployment despite primary ALB being healthy.

Root Cause: Failover criteria too sensitive - 503 errors during rolling deployment triggered unnecessary failover.

Resolution: Excluded 503 from failover criteria, implemented threshold-based failover (3 consecutive errors).

Skills Demonstrated: CloudFront configuration, origin groups, deployment analysis, monitoring

Preventative Actions: Deployment-aware monitoring, Lambda automation for failover during deployments

CS-003: CI/CD Pipeline Deployment Failure

Priority	High
Service	CodePipeline, CodeDeploy, IAM
Status	Resolved
Resolution Time	1 hour 45 minutes

Problem: Pipeline succeeded in build stage but failed during deployment with AccessDeniedException.

Root Cause: CodeDeploy service role missing ec2:CreateTags permission required for deployment tracking.

Resolution: Created custom IAM policy with ec2:CreateTags, attached to CodeDeploy service role with least privilege.

Skills Demonstrated: CI/CD debugging, IAM troubleshooting, permission analysis, policy simulation

Preventative Actions: Permission validation script, automated IAM checks in pipeline, quarterly policy reviews

Skills Demonstrated

Category	Skills
Technical Troubleshooting	VPC networking, security groups, CloudFront origin groups, CI/CD pipelines, IAM permissions
Investigation Methods	Log analysis (CloudWatch), metrics interpretation, systematic debugging, policy simulation
Root Cause Analysis	Identifying underlying issues vs. symptoms, correlation analysis, timeline reconstruction
Problem Resolution	Configuration fixes, permission updates, threshold tuning, automation implementation
Professional Skills	Clear documentation, customer communication, preventative thinking, time management
AWS Services	EC2, VPC, Security Groups, CloudFront, CodePipeline, CodeBuild, CodeDeploy, IAM, CloudWatch